

## HALLEY STATION, Antarctica

WMO: 890220

Lat: 75.607S

Lon: 26.208W

Elev: 98

StdP: 14.64

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -49.8      | -45.5 | -56.7                         | 0.2 | -49.2 | -52.6 | 0.2 | -45.3 | 52.6                  | -2.0 | 47.9 | -1.5 | 6.3                   | 90   | 1.358 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.4                   | 31.8            | 30.4 | 30.5 | 29.3 | 29.3 | 28.2 | 30.7                | 31.4 | 29.6 | 30.3 | 28.3 | 29.1 | 13.6                 | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 30.1                            | 24.3 | 30.8 | 28.7 | 22.7 | 29.9 | 27.2 | 21.2 | 28.8 | 11.3          | 31.5 | 10.9 | 30.5 | 10.3 | 29.3 | 38.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 44.5          | 39.3 | 33.4 | DB                         | -55.9 | 34.7               | 5.0 | 2.0                                                | -59.5 | 36.2       | -62.4 | 37.4       | -65.2 | 38.5       | -68.9 | 40.0 | (4) |
| (5)               |      |      | WB                         | -55.9 | 32.1               | 5.0 | 1.3                                                | -59.5 | 33.1       | -62.4 | 33.9       | -65.2 | 34.6       | -68.8 | 35.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | -3.2   | 22.5  | 11.8  | 0.2   | -10.5 | -13.6 | -17.5 | -20.2 | -20.0 | -16.4 | -5.9  | 11.0  | 21.5  |
|                                                                               | DBStd       | 19.12  | 4.74  | 7.90  | 11.44 | 13.54 | 13.11 | 14.99 | 13.37 | 13.74 | 13.61 | 10.79 | 7.61  | 4.81  |
|                                                                               | HDD50       | 19405  | 851   | 1071  | 1545  | 1814  | 1971  | 2026  | 2176  | 2170  | 1993  | 1733  | 1170  | 885   |
|                                                                               | HDD65       | 24880  | 1316  | 1491  | 2010  | 2264  | 2436  | 2476  | 2641  | 2635  | 2443  | 2198  | 1620  | 1350  |
|                                                                               | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 13.0   | 10.3  | 11.5  | 12.9  | 12.8  | 12.9  | 14.4  | 13.4  | 13.2  | 14.6  | 14.4  | 14.6  | 11.4  |
| Precipitation                                                                 | PrecAvg     | 11.6   | 0.7   | 0.8   | 0.9   | 0.8   | 0.9   | 1.0   | 1.0   | 1.1   | 1.3   | 1.0   | 1.3   | 0.8   |
|                                                                               | PrecMax     | 18.1   | 1.5   | 2.2   | 2.0   | 1.6   | 2.3   | 2.9   | 2.1   | 2.1   | 3.1   | 3.6   | 4.3   | 2.3   |
|                                                                               | PrecMin     | 7.4    | 0.2   | 0.1   | 0.3   | 0.1   | 0.2   | 0.3   | 0.2   | 0.1   | 0.2   | 0.1   | 0.2   | 0.2   |
|                                                                               | PrecStd     | 2.6    | 0.4   | 0.4   | 0.4   | 0.4   | 0.6   | 0.7   | 0.5   | 0.5   | 0.7   | 0.7   | 0.8   | 0.5   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 33.8  | 30.6  | 28.1  | 23.3  | 24.7  | 19.3  | 12.1  | 18.9  | 17.5  | 20.7  | 28.9  | 33.0  |
|                                                                               |             | MCWB   | 31.8  | 29.9  | 27.9  | 22.7  | 24.5  | 18.3  | 11.6  | 18.7  | 16.8  | 20.1  | 27.6  | 30.8  |
|                                                                               | 2%          | DB     | 32.2  | 28.6  | 24.0  | 19.5  | 15.8  | 14.6  | 8.9   | 11.8  | 13.1  | 17.7  | 26.2  | 31.5  |
|                                                                               |             | MCWB   | 30.8  | 27.9  | 23.6  | 19.1  | 15.2  | 13.9  | 8.4   | 11.1  | 12.6  | 17.1  | 25.2  | 30.0  |
|                                                                               | 5%          | DB     | 31.0  | 26.4  | 21.0  | 16.2  | 9.8   | 10.1  | 4.5   | 5.9   | 9.5   | 14.8  | 24.4  | 30.3  |
|                                                                               |             | MCWB   | 29.9  | 25.7  | 20.5  | 15.6  | 9.5   | 9.6   | 4.1   | 5.4   | 9.0   | 14.2  | 23.5  | 29.0  |
|                                                                               | 10%         | DB     | 29.8  | 24.1  | 17.5  | 12.0  | 5.6   | 5.2   | -0.2  | 0.9   | 4.7   | 11.3  | 22.5  | 29.0  |
|                                                                               |             | MCWB   | 28.8  | 23.3  | 16.8  | 11.5  | 5.1   | 4.7   | -0.6  | 0.5   | 4.2   | 10.7  | 21.7  | 27.7  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 32.4  | 30.3  | 28.0  | 22.9  | 24.5  | 18.7  | 11.6  | 18.8  | 17.1  | 20.3  | 27.9  | 31.7  |
|                                                                               |             | MCDB   | 32.9  | 30.5  | 28.0  | 23.2  | 24.7  | 19.1  | 11.8  | 19.0  | 17.3  | 20.5  | 28.6  | 32.2  |
|                                                                               | 2%          | WB     | 31.2  | 28.0  | 23.8  | 19.2  | 15.3  | 14.1  | 8.4   | 11.5  | 12.6  | 17.1  | 25.5  | 30.4  |
|                                                                               |             | MCDB   | 31.8  | 28.4  | 24.0  | 19.5  | 15.8  | 14.8  | 8.8   | 12.1  | 12.9  | 17.5  | 26.1  | 31.2  |
|                                                                               | 5%          | WB     | 30.2  | 25.8  | 20.6  | 15.7  | 9.4   | 9.6   | 4.1   | 5.4   | 9.2   | 14.3  | 23.5  | 29.2  |
|                                                                               |             | MCDB   | 30.8  | 26.4  | 20.9  | 16.1  | 9.7   | 10.0  | 4.5   | 6.0   | 9.6   | 14.8  | 24.2  | 30.1  |
|                                                                               | 10%         | WB     | 29.0  | 23.4  | 16.9  | 11.5  | 5.1   | 4.8   | -0.5  | 0.5   | 4.2   | 10.7  | 21.7  | 27.8  |
|                                                                               |             | MCDB   | 29.8  | 24.0  | 17.4  | 11.9  | 5.5   | 5.1   | -0.2  | 0.8   | 4.6   | 11.2  | 22.4  | 28.8  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.4  | 13.5  | 14.5  | 15.5  | 15.4  | 15.1  | 15.1  | 15.6  | 15.1  | 15.8  | 13.1  | 10.7  |
|                                                                               |             | MCDBR  | 10.4  | 11.5  | 12.3  | 13.4  | 15.3  | 14.9  | 15.9  | 15.4  | 12.8  | 14.6  | 12.0  | 11.0  |
|                                                                               |             | MCWBR  | 9.4   | 11.1  | 12.2  | 13.4  | 15.3  | 14.9  | 15.8  | 15.5  | 12.8  | 14.4  | 11.4  | 10.0  |
|                                                                               | 5% WB       | MCDBR  | 9.2   | 10.7  | 12.0  | 12.8  | 15.4  | 14.8  | 15.9  | 15.5  | 13.0  | 14.4  | 11.8  | 10.6  |
|                                                                               |             | MCWBR  | 8.4   | 10.4  | 12.2  | 12.9  | 15.4  | 14.9  | 15.8  | 15.6  | 13.0  | 14.3  | 11.2  | 9.7   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.258  | 0.242 | 0.202 | 0.124 | N/A   | N/A   | N/A   | N/A   | 0.124 | 0.191 | 0.216 | 0.235 | 0.254 |
|                                                                               | taud        | 2.332  | 2.339 | 2.262 | 1.913 | N/A   | N/A   | N/A   | N/A   | 1.881 | 2.180 | 2.246 | 2.289 | 2.315 |
|                                                                               | Ebn at Noon | 303    | 283   | 246   | 99    | 0     | 0     | 0     | 0     | 89    | 244   | 292   | 311   | 311   |
|                                                                               | Edh at Noon | 30     | 24    | 16    | 3     | 0     | 0     | 0     | 0     | 3     | 17    | 26    | 31    | 32    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2188   | 1356  | 543   | 71    | 0     | 0     | 0     | 0     | 19    | 317   | 1109  | 2117  | 2551  |
|                                                                               | RadStd      | 117    | 62    | 25    | 6     | 0     | 0     | 0     | 0     | 1     | 20    | 50    | 65    | 99    |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page